Frequency Tables

Frequency: To organise data, we create a **frequency table**. The frequency table shows how often something happens. To tally means to count.

Ungrouped Data

In the following table the results are displayed of a survey in which 25 adults were asked how big their families were:

Number of children	Tally	Frequency
1	//	2
2	~~###~~ ////	9
3	~~###~~	5
4	////	4
5	///	3
6	//	2
	Total	25

From the example, we can see that the **total number of observations** is 25. This is also referred to as the **total frequency.**

Activity 1

Thabo did a survey concerning how many days learners were absent from school. He noticed that nobody was absent for more than 9 days. He recorded the following data:

2	3	9	0	2
4	0	4	1	8
4	6	5	1	7
4	3	9	2	9
6	6	3	4	1
0	4	5	0	3
8	2	3	3	4

Complete the frequency table to make sense of the given data and answer the questions that follow:

Days Absent	Tally	Frequency
0	IIII	4
1	III	3
2	IIII	4
3	~~IIII~~ I	6
4	~~IIII~~ II	7
5	II	2
6	III	3
7	I	1
8	II	2
9	III	3
	Total	35

a. How many learners were in the survey? 35

b. How many learners were absent for 3 days or less during the term? 17

c. How many learners were absent for 4 days of the term? 7

d. What percentage of learners were absent for more than 4 days? 11 learners

$$\frac{11}{35} \times 100 = 31,43\%$$

Grouped Data

Grouped Data: When the range of a set of data is large, it is helpful to divide the data into intervals of equal size.

A complete list of grouped frequencies is called a **grouped frequency distribution.**

Class Intervals: A group or a category is called a class interval. The **width** or size of the class interval will depend on the data collected.

Example:

The following table shows the number of kilometres, learners of the Phokani high school live away from the school. The data was grouped in intervals to make it easier to depict.

Number of Kilometres	Frequency
under 5	55
$5 \leq x < 10$	30
$10 \leq x < 15$	28
$15 \leq x < 20$	23
$20 \leq x < 25$	10
$25 \leq x < 30$	0
$30 \leq x < 50$	4



Note:
No value falls in more than ONE class. In the interval $15 \leq x < 20$ we refer to 15 as the **lower class limit** and 20 as the **upper class limit.**

$$15 \leq x < 20$$

Lower class limit **Upper class limit**

Activity 2

Consider the table and answer the following:

Number of Kilometres	Frequency
under 5	55
$5 \leq x < 10$	30
$10 \leq x < 15$	28
$15 \leq x < 20$	23
$20 \leq x < 25$	10
$25 \leq x < 30$	0
$30 \leq x < 50$	4

1) a. How many people in the survey were asked what distance they stay away from school? 150

b. How far away do most learners stay? under 5 km

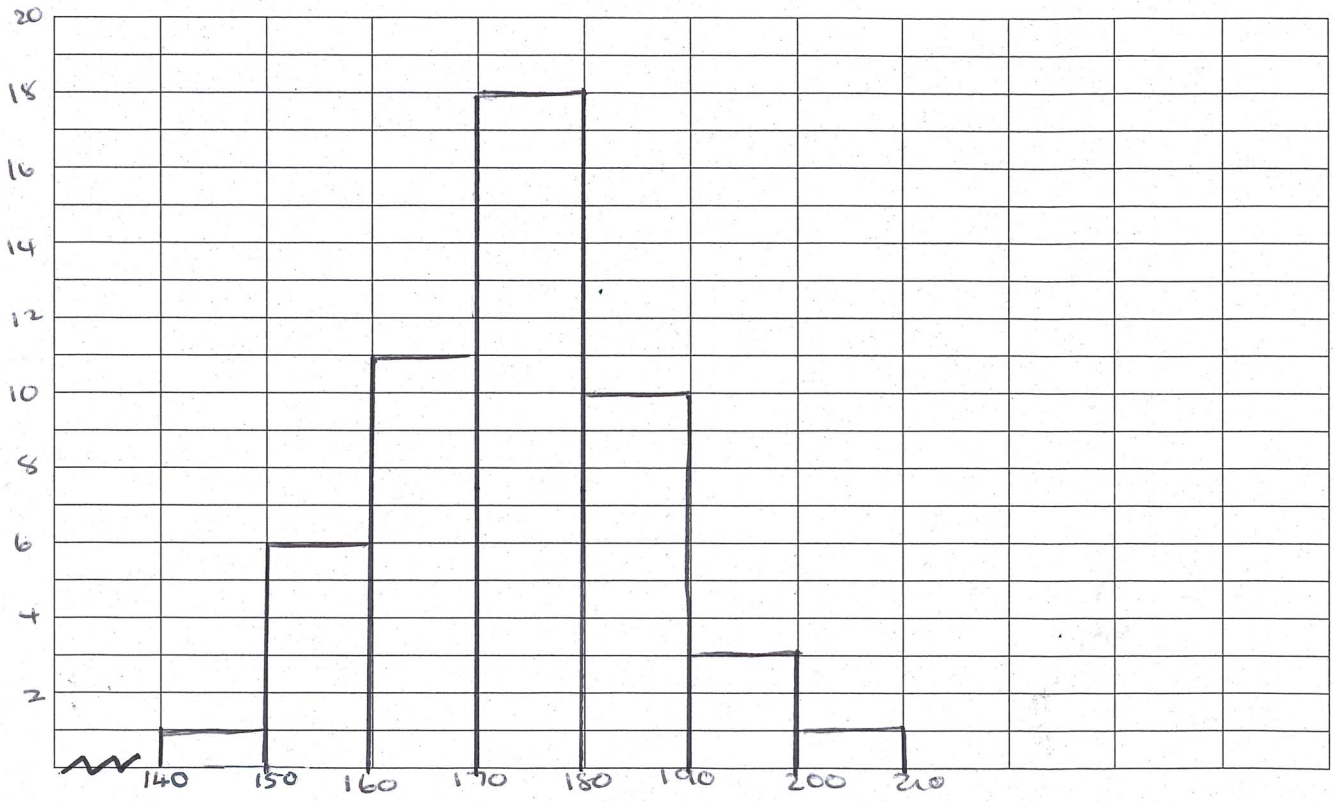
c. How many people stay more than 20 km away from school? 14

d. What percentage of learners travel more than 60 km per day to school and back? $\frac{4}{150} \times 100 = 2,67\%$

e. You want to start a bus service to the school. Would you consider this to be a good option in the area of Phokani Secondary School?
Give a reason for your answer. Yes. 95 learners live 5 - 50 km away and will need transport.

2) The heights of 50 adults are given in the table. Draw a histogram to display the data:

Height in cm	Number of people (frequency)
$140 < h \leq 150$	1
$150 < h \leq 160$	6
$160 < h \leq 170$	11
$170 < h \leq 180$	18
$180 < h \leq 190$	10
$190 < h \leq 200$	3
$200 < h \leq 210$	1



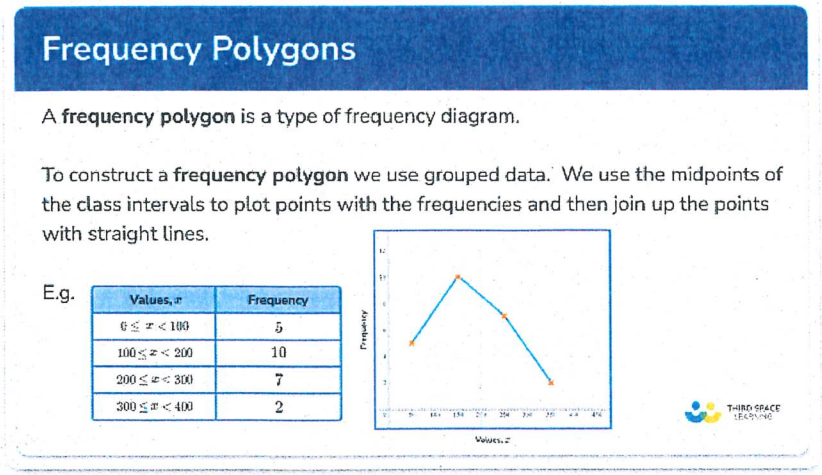
a. Which is the modal class? $170 < h \leq 180$

b. How many adults were included in the survey? 50

c. How many adults have a height of more than 170cm? 32

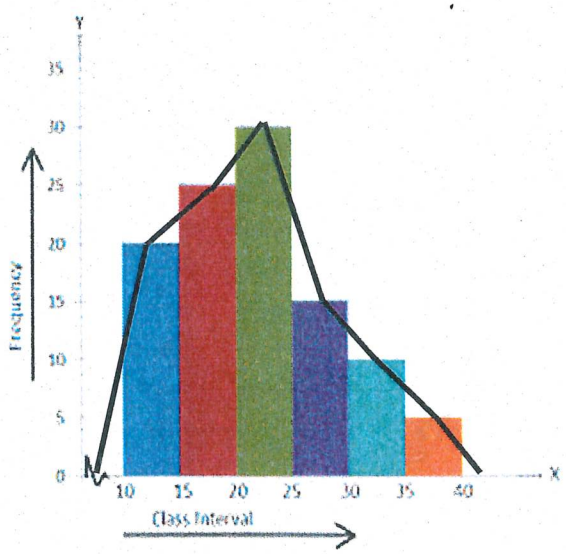
Frequency Polygons

Both the histogram and the frequency polygon are frequency graphs.



How to draw a frequency polygon by using a histogram

1. Draw the Histogram of the data given
2. Mark the midpoints on the top of every bar with a dot
3. Join the dots together, in order from left to right with ***straight lines***
4. Remember to start the first dot and end the last dot on an imaginary class at the beginning and end of the histogram with a frequency of 0.



How to draw a frequency polygon without first drawing a histogram

1. Find the midpoints of every class interval
2. Plot the midpoint of the class interval against its frequency
3. Join all the plotted points in order for each class
4. Extra classes with zero frequency are added at the beginning and end of the frequency distribution. These points must be plotted and all dots joined with straight lines.

It is important to note that the Frequency polygon is a closed two dimensional shape made from the lines of the graph and the horizontal axis.

Finding the midpoint for continuous data

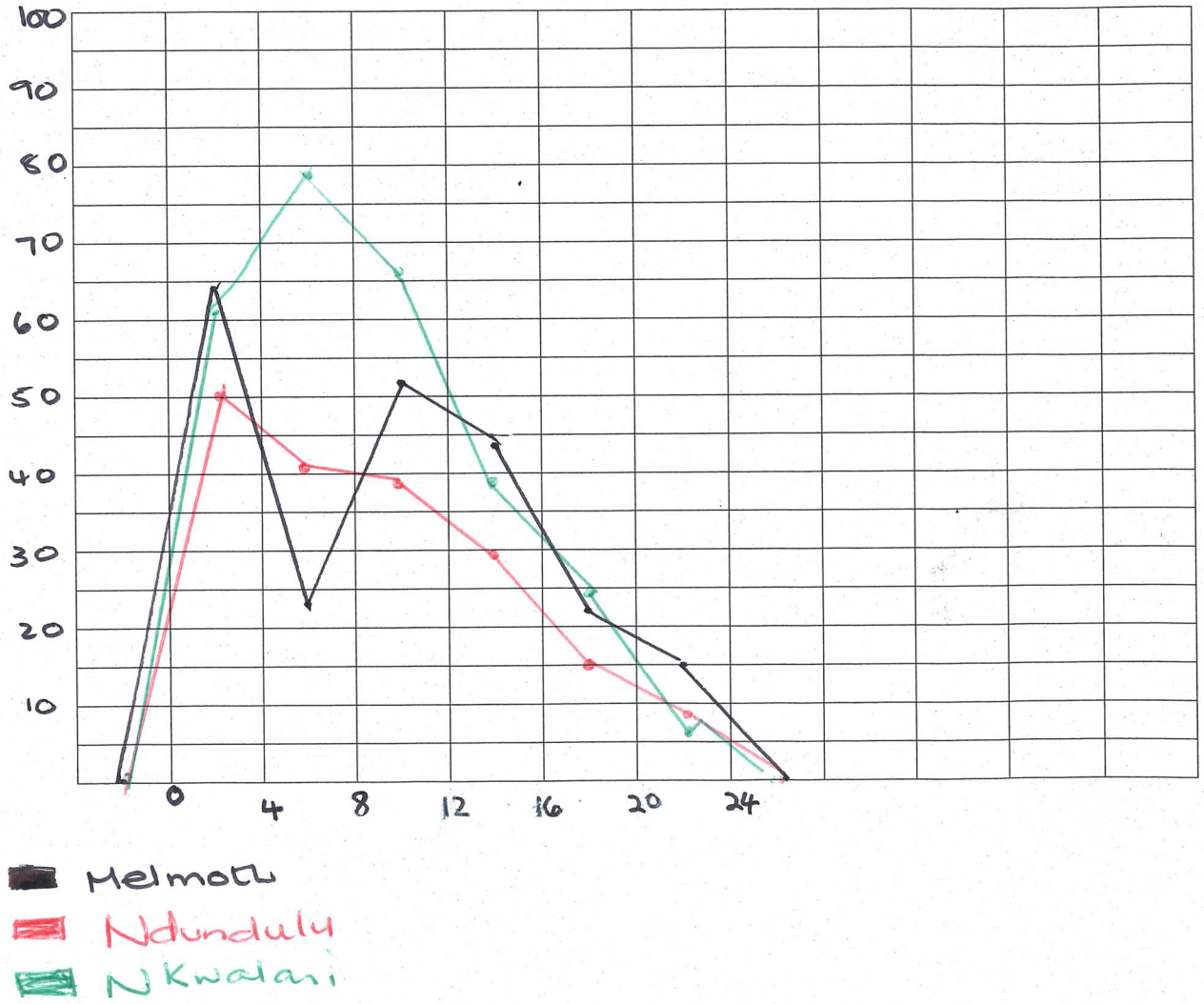
½ (lower class boundary + upper class boundary)

Activity 3

Three schools near Richard's Bay decided to do a survey to see how far their learners walk to school. The findings of the schools are reflected in the following table:

	Distance in kilometres from school					
School	$0 \leq x < 4$	$4 \leq x < 8$	$8 \leq x < 12$	$12 \leq x < 16$	$16 \leq x < 20$	$20 \leq x < 24$
Melmoth Secondary	64	23	52	43	22	15
Ndundulu Primary	55	41	38	29	15	8
Nkwalani Secondary	61	78	66	38	24	7

Draw the frequency polygons of the given information in different colours on the same graph below then answer the questions that follow: **Remember to label your axes.**



a. Compare the pattern at the different schools.

(1) Melmoth - random patter

(2) Ndundulu - most learners that walk Stay within 4km of School.

(3) Nkwalani - most learners walk between 4 + 8 km to school.

b. Find the most common distance in each school that the learners walk.

(1) Melmoth 0 → 4

(2) Ndundulu 0 → 4

(3) Nkwalani 4 → 8.

c. How many learners in each school walk more than 8km per day?

(1) Melmoth = 132

(2) Ndundulu = 90

(3) Nkwalani = 135

d. If you owned a bus service, which school would you rather offer your service to and why? Nkwalani - more learners would make use of it as largest no of learners living far from school.

Activity 4

1) Go back to Activity 1 and calculate the mean number of days learners were absent from school. $\bar{x} = 3,85$

2) Can you find the median? $17,5 \approx 18$ th position $\therefore$ media is 4 (days)

3) What is the modal number of days?.

4

Grouped data

With grouped data we do not have the actual values of every individual case, therefore we cannot find the true value of the mean, but we can *estimate the value* and this is how we go about it.

Say we did a survey of the shoe sizes of grade 10 girls:

Size of shoes	Frequency
$4 \leq x < 5$	32
$5 \leq x < 6$	28
$6 \leq x < 7$	12
$7 \leq x < 8$	1

The first we have to do is to find out what is the *midpoint* of every class:
The midpoint of a class is determined in the following way:

Consider the class $4 \leq x < 5$, the midpoint between 4 and 5, is $\frac{4+5}{2} = 4,5$

So we complete the table:

Size of shoes	Frequency	Midpoint
$4 \leq x < 5$	32	4,5
$5 \leq x < 6$	28	5,5
$6 \leq x < 7$	12	6,5
$7 \leq x < 8$	1	7,5

Now we find the frequency multiplied by the midpoint:

Size of shoes	Frequency (f)	Midpoint (m)	Frequency X Midpoint (f X m)
$4 \leq x < 5$	32	4,5	32 X 4,5 = 144
$5 \leq x < 6$	28	5,5	28 X 5,5 = 154
$6 \leq x < 7$	12	6,5	12 X 6,5 = 78
$7 \leq x < 8$	1	7,5	1 X 7,5 = 7,5
	Total = 73		*Total* = 383,5

$$\textit{Estimated mean is}: \frac{\text{Total of the frequency} \times \text{midpoint}}{\text{Frequency}}$$

$$= \frac{383,5}{73} = 5,3 \rightarrow$$

Therefore the estimated mean is: 5,3

USING YOUR CALCULATOR TO FIND THE ESTIMATED MEAN FOR A FREQUENCY TABLE (Grouped Data)

★ Same as for ungrouped, but use midpoint for x-value ★

Activity 5

Use your calculator to find the estimated mean for the example above.

Mode of Grouped Data:

Consider the following information about the T-shirts sold at a shop:

Sizes	Frequency
$30 \leq x \leq 34$	12
$36 \leq x \leq 40$	28
$42 \leq x \leq 46$	9

***Questions:**

In which class interval will you find the most T-shirts sold? $36 \leq x \leq 40$
[In other words, which class interval will have the highest frequency?]

Can we find the median value? $\frac{49}{2} = 24,5 \approx 25$th position.
∴ median would lie in $36 \leq x \leq 40$ interval.

Activity 6

1. In the *Business Report* of *The Star* of 19 May 2008, this following information was reported in the *Fast Facts-* column:
"In Africa, 42 out of the 54 countries in Africa have **median** ages below age 20."

   a. What do you understand by this statement?
   Half of the population is younger than 20 years of age.

   b. What are the implications of this statistic?
   A large number of the population depends economically on relatively few older people.

2. From the entry forms of a half marathon, a judge counted the ages of female runners within certain age groups. The results are recorded in the table below:

a. Complete the table with class intervals of 10 years, starting with values of $20 \leq x < 30$, etc. of the female runners.

Female runners' ages	Frequency	Midpoint	Midpoint X Frequency
$20 \leq a < 30$	8	25	200
$30 \leq a < 40$	12	35	420
$40 \leq a < 50$	24	45	1080
$50 \leq a < 60$	9	55	495
$60 \leq a < 70$	5	65	325
	Total = 58		Total = 2520

b. From the table determine:

(i) the modal class interval $40 \leq a < 50$

(ii) the estimated mean age 43,45

(iii) the median age interval $\frac{58}{2}$ = 29th position.

$\therefore 40 \leq a < 50$

3. The following table shows the marks obtained by 40 learners in a mathematics test.

Marks:
12, 18, 25, 34, 40, 28, 33, 7, 10, 22, 19, 35, 42, 29, 0, 5, 37, 41, 13, 8, 30,
45, 21, 26, 38, 14, 12, 39, 44, 32, 16, 6, 23, 27, 11, 17, 36, 48, 31, 15

1. Fill in the frequency column in the table below using the data above.
2. Fill in the midpoint of each interval on the table.

Marks (out of 50)	Midpoint	Frequency
0 – 9	$4\frac{1}{2}$	5
10 – 19	$14\frac{1}{2}$	11
20 – 29	$24\frac{1}{2}$	8
30 – 39	$34\frac{1}{2}$	10
40 – 49	$44\frac{1}{2}$	6

Using your calculator, find the estimated mean of the data.

$\bar{x} = 24,75$

Activity 7

1. The speed in kilometres per hour was recorded for vehicles on the N3 between Durban and Johannesburg:

110	129	143	133	136	127	122	156	134	111	68
44	143	126	131	160	146	121	105	109	115	76

Find:

a. the minimum, maximum values and the range _____

min: 44 max: 160 range: 160 - 44 = 116 km/L.

b. the median $Q_2 = 126,5$ km/L

c. the lower quartile $Q_1 = 110$ km/L

d. the upper quartile $Q_3 = 136$ km/L

e. the interquartile range $136 - 110 = 26$ km/L

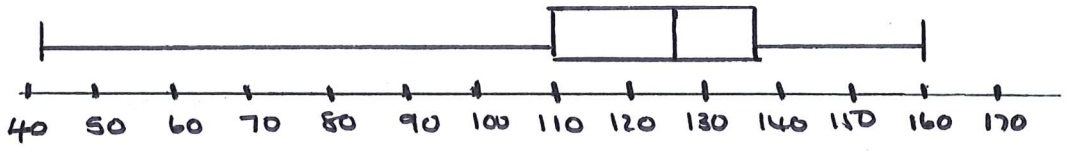
f. the semi-interquartile range of speed that the motorists travelled. _____

$26 \div 2 = 13$ km/L

g. If the speed limit is 120 km per hour, what can you conclude? _____

More than half the motorists drove above the speed limit.

h. Draw the box and whisker plot for the data



2. The *Talk-Talk* cell-phone company recorded the time learners between the age of 14 and 18 spend on their cell phones.

Number of minutes	Number of calls	Cumulative Frequency
$0 \leq t < 3$	33	33
$3 \leq t < 6$	66	99
$6 \leq t < 9$	42	141
$9 \leq t < 12$	25	166
$12 \leq t < 15$	14	180

Cumulative frequency is the running total of frequencies in a frequency distribution. Complete the cumulative frequency of the number of calls and find:

a. The number of calls made in the survey. 180

b. Which interval contains the lower quartile? 45 values below $3 \leq t < 6$

c. Which interval contains the upper quartile? 45 values above $9 \leq t < 12$

d. What percent of the learners spend more than 3 minutes on their cell phones?

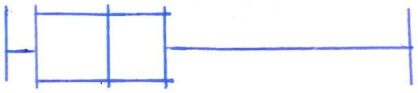
$\frac{147}{180} \times 100 = 81,67\%$

3. The box-and-whisker plot helps us to compare two sets of data easily. Consider the results of a sample of learners in a given Maths test and an English test. Both tests are out of 25.

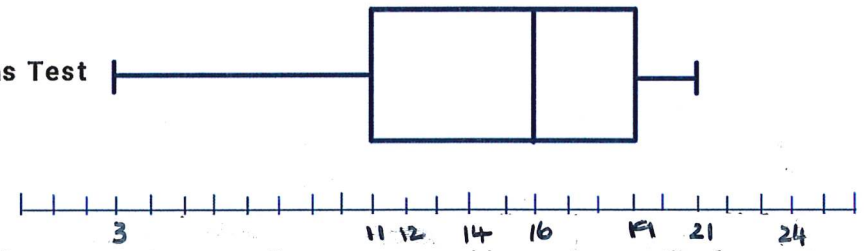
Mathematics	3	8	11	11	13	15	16	18	18	18	20	20	21
English	11	12	12	12	13	14	14	15	15	16	16	19	24

a. Work out the five- number summary of both tests and draw the box plot of the English test above the Maths test results, using the same number line below:

English Test:



Maths Test



a. How many test results are covered in each quartile for each subject? 3

b. What does the inter-quartile ranges of the two different subjects tell us?

Compare the lengths of the two boxes. The middle 50% in English are closer together (IQR of 4) than in Maths (IQR of 8)

c. Are there more pieces of data in the first than in the last quartile of the Maths test?

Explain.

No, all quartiles contain the same amount of data.

d. What does the difference in length of the whiskers indicate? Give the range of values for each subject's whiskers. Maths marks are spread wider (21 - 3 = 18) than English marks (24 - 11 = 13)
English max mark of 24 is an outlier.

e. What does the position of the median in each test tell us?
Half the English class scored above/below 14/25
Half the Maths class scored above/below 16/25